

Week 12: Lean macros

Functional Programming in Lean

Learning goals

- ▶ Syntax quotations
- ▶ Antiquotations
- ▶ Term and tactic macros

Core concepts

- ▶ Macro expansion: Syntax to Syntax
- ▶ Quotation: parsed syntax as data
- ▶ Antiquotation: captured syntax insertion

Lean example

```
syntax "twice(" term ")" : term
macro_rules
  | `(twice($x)) => `($x + $x)
#eval twice(21)
```

Lean details

- ▶ `twice`: term macro
- ▶ `close_simple`: tactic macro
- ▶ Kernel checking after elaboration

Live coding

- ▶ `twice` expansion
- ▶ `close_simple` expansion
- ▶ Unsolved goal after tactic failure

Check

Role of the kernel after macro expansion

Exercise

- ▶ `thrice` macro
- ▶ A goal outside `close_simple`